Network Control Protocol

(NCP)

What is NCP?

NCP stands for Network Control Protocol.

- It is used in PPP (Point-to-Point Protocol) after the LCP (Link Control Protocol) has established the link.

- Its main job is to allow multiple network layer protocols to operate over the same PPP link.

Purpose of NCP

- To establish and configure network layer protocols over PPP
- To encapsulate different types of packets (e.g., IP, IPX)
- To negotiate protocol-specific options like IP addresses
- To terminate network protocol communication when needed

How NCP Works

- After LCP completes link setup, NCP begins
- Each network layer protocol has its own NCP (e.g., IPCP for IP, IPXCP for IPX)
- NCP exchanges configuration requests and responses between devices
- Once NCP succeeds, the link is ready for actual data transfer

Examples of NCPs

NCP	Network Layer Protocol	Short Description
IPCP	Internet Protocol Control Protocol	Configures IP addresses and options for data transfer over PPP.
IPXCP	Internetwork Packet Exchange Control Protocol	Configures IPX addresses and settings, mainly used in old Novell networks.
ATCP	AppleTalk Protocol	Configures AppleTalk addresses for Apple networking over PPP.
IPV6CP	IPv6 Control Protocol	It helps in the configuration of the IPv6 addresses it also enables and disables IP protocol modules over PPP.

Functions of NCP

- Assigning addresses for the network layer protocol
- Negotiating options specific to the protocol
- Starting or stopping communication of a specific protocol
- Supporting multiple protocols simultaneously

Summary

- NCP comes after LCP in PPP
- Allows multiple network layer protocols over one PPP link
- Each protocol has a specific NCP
- Prepares the link for actual data transfer